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MH1200 Linear Algebra I

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1. Expressing Two Row-Equivalent Matrices as Products of One Another

Given two row-equivalent matrices A and B, there exists an invertible matrix Q such that:

$$A = QB$$

To find this matrix Q, one can reduce both A and B to RREF. Note down the elementary row operations used, and convert the sequence to a matrix product.

Then, you will have some equation of the form:

$$P_n P_{n-1} \cdots P_1 A = Q_m Q_{m-1} \cdots Q_1 B$$

Where all P_i and Q_j are elementary matrices (and are thus invertible).

Then, we have:

$$Q = P_1^{-1} P_2^{-1} \cdots P_n^{-1} Q_m Q_{m-1} \cdots Q_1 B$$

2. Characterizations of Invertibility

What is a **"characterization"**?

characterization simple properties of a matrix different from invertibiliity but are equivalent to it.

Theorem | Characterizations of Invertibility

Theorem 1

Given a square matrix A, the following statements are equivalent:

1. A is invertible.
2. The only solution of $A\vec{x} = \vec{0}$ is the trivial solution $\vec{x} = \vec{0}$.
3. The reduced-row echelon form (RREF) of A is the identity matrix.
4. A can be expressed as the product of elementary matrices.

How do you prove it?

We'll **hit a home run** (darn such a good analogy).

We will prove them in a circular manner.

Proof | Characterizations of Invertibility

Proof 2

We want to prove that $1 \to 2$, $2 \to 3$, $3 \to 4$, and $4 \to 1$.

Suppose A is invertible. We want to show that $A\vec{x} = \vec{0}$ only has $\vec{x} = \vec{0}$ as its solution.

Suppose ftsoc that there exists $\vec{x} \neq \vec{0}$ such that

$$A\vec{x} = \vec{0}$$

Since A is invertible, there exists a matrix A^{-1} such that:

$$A^{-1}A = I$$

Then:

$$A\vec{x}=\vec{0}$$

$$A^{-1}A\vec{x}=A^{-1}\vec{0}$$

$$\vec{x}=A^{-1}\vec{0}$$

$$\vec{x}=\vec{0}$$

Contradiction. Thus, $1 \to 2$.

Next, suppose $A\vec{x} = \vec{0}$ only has $\vec{x} = \vec{0}$ as its solution. We want to show that the reduced row echelon form (RREF) of A is the identity matrix.

Suppose ftsoc that the RREF of A is not the identity matrix. Then, the RREF of A has at least one column with a nonleading entry. Further, $A\vec{x} = \vec{0}$ has at least one solution, so the equation $A\vec{x} = \vec{0}$ corresponding to the augmented matrix with at least one column with no leading entries $\left(A \mid \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}\right)$ has infinite solutions, contradiction. Thus, $2 \to 3$.

Next, suppose the RREF of A is the identity matrix. We want to show that A is the product of some elementary matrices. Gauss-Jordan elimination gives us steps to reduce A to the identity matrix. Therefore, there are elementary matrices E_i such that:

$$E_n E_{n-1} E_{n-2} \cdots E_1 A = I$$

Since all elementary matrices are invertible,

$$A = E_1^{-1} E_2^{-1} E_3^{-1} \cdots E_n^{-1}$$

Since E_i^{-1} is elementary, A is indeed the product of elementary matrices. Thus, $3 \to 4$.

Finally, suppose A can be expressed as the product of elementary matrices. We want to show that it is invertible. Let $(E_i)_{i=1}^n$ be some sequence of elementary matrices such that:

$$A = E_n E_{n-1} E_{n-2} \cdots E_1$$

Then,

$$E_1^{-1} E_2^{-1} E_3^{-1} \cdots E_n^{-1} A = I$$

Thus,

$$E_1^{-1} E_2^{-1} E_3^{-1} \cdots E_n^{-1}$$

is the inverse of A, implying that A is invertible. Thus, $4 \to 1$.